

HARINI ANAND

Bangalore, India

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Education

PES University

August 2021 – August 2025

Bachelor of Technology in Computer Science and Engineering

Bangalore, Karnataka

Specialization in Machine Intelligence and Data Science

Relevant Coursework

- Data Structures
- Artificial Intelligence & ML
- Computer Networks
- Operating Systems
- Cloud Computing
- NLP
- Kubernetes

Experience

IBM

August 2025 – Present

Software Developer

Bangalore, Karnataka

- Achieved a **90% cost reduction** in cloud file storage by optimizing configurations, reducing costs from \$820 to \$70.
- Selected to **lead the Speaking & Authoring Working Group** under the ISL-wide **Core-X** initiative, driving technical storytelling and innovation culture across IBM.
- Contributing to the **Agentic AI patent project**, exploring novel frameworks for intelligent automation and cognitive decision systems.
- Collaborating on the **OpenSearch acquisition** tech preview, focusing on integration for industry and enterprise-scale use cases.
- Worked on the **FedRAMP compliance project**, strengthening cloud security posture and implementing required federal controls.
- Developed multiple **PoCs using LangFlow and Groq**, showcasing initiative and problem-solving through AI-driven automation workflows.

UIUC, University of Illinois Urbana-Champaign

June 2025 – Present

Software Engineering Researcher

Remote

- * Selected to join a remote research cohort under Prof. Darko Marinov (MIR Group, UIUC) for the UIUC+ Summer Undergraduate Research in Software Engineering Program.
- * Ranked among the top **15** researchers out of **1,300+** applicants through multi-stage selection.
- * Researched fairness testing in machine learning systems, extending symbolic execution and local explainability techniques to multimodal models; led a peer reading group and synthesized findings from recent ICSE and FSE 2025 papers on fairness, bias mitigation, and intersectional testing.
- * Contributed to research on automated program repair (APR) by analyzing repair pipelines (e.g., Agentless, TARGET, SEER), identifying bug-inducing commits using the SZZ algorithm, curating verified datasets from open-source repositories and ISSTA 2025 papers, and improving collaborative research infrastructure.

IBM

Jan 2025 – July 2025

Software Developer Intern

Bangalore, Karnataka

- * Working on the **Data and AI** domain to build, develop and implement data architecture solutions for Watsonx data.
- * Contributing to the Cloud Services team to build internal CLI tooling using **Go and Kubernetes API Machinery**
- * Designed and implemented custom commands for interacting with CRDs across multiple ROKS, and Azure clusters
- * **Speaker, KubeCon India 2025** - Presented on Kubernetes developer tooling at Watsonx Data; only intern accepted as a speaker

Niramai Health Analytix

Jan 2024 – July 2024

Software Developer Intern

Bangalore, Karnataka

- * Collaborated with the product management team to design, **develop** and build an end-to-end women's health application with **20 features** for **breast cancer screening** using Flutter for the cross-platform mobile development in **5 months**.
- * Integrated **self-examination & tracking tools** to enhance **predictive analytics**, enabling users to track health metrics using **Firebase** for backend services, with **real-time database management, authentication, and cloud storage**.
- * Implemented **user profile creation workflow**, capturing detailed user information and storing it securely in the **PII table**, generating unique Patient IDs for self-breast exams, resulting in **40%** increased user engagement and compliance.

Indian Institute of Technology, Hyderabad

June 2023 - December 2023

Machine Learning Research Intern

Hyderabad

- * Utilized **PySCENIC** to analyze single-cell RNA-seq data, and work on **predicting gene regulatory networks**.
- * Integrated **machine learning algorithms and decision trees** for predictive modeling of gene regulatory interactions.
- * Selected as one of **40** students nationwide for the 15 day Summer School hosted by **ACM India & IIT Hyderabad**.

Projects

Project ROOTS | *Flutter, Firebase, Figma* | [Link](#)

January 2022 - December 2022

- * Developed a Flutter app aimed at reducing the onset of dementia among senior citizens using personalized cognitive algorithms and positively impacted over **350** users across the state to provide support for memory reinforcement.
- * Won the **Cisco ThingQbator Cohort 5** national competition and received a **5 Lakh seed grant** for the project.

Restaurant Recommendation System | *Python, Data Science, Data Analysis* | [Link](#)

January 2023 - March 2023

- * Secured an educational grant of **1000 USD** from **Google KaggleX** and participated in a 4-month global technical mentorship program to develop an end-to-end pipeline restaurant recommendation system, optimizing suggestions based on **cost, ratings, cuisine, and location** to address issues faced by students and culinary entrepreneurs by factoring in multiple constraints and utilizing **real-time data collection** mechanisms to capture the **latest** restaurant information.
- * Used **K Means Clustering, TF-IDF, Collaborative Filtering & Cosine Similarity** to segment data, **feature engineering** processes to improve **model performance** and **EDA** to gain insights into **restaurant attributes**.

Technical Skills

Languages/Softwares: Python, SQL, C, HTML5/CSS, JavaScript, R, MATLAB, Bash, Agile Development, DevOps
Developer Tools: Jupyter, VS Code, Google Cloud Platform, Colab, Kubernetes, Anaconda, JIRA, Jenkins, Git, pip, npm
Technologies/Frameworks: TensorFlow, PyTorch, Scikit-learn, Pandas, GitHub, Docker, HuggingFace, NLTK, Keras

Achievements

Harvard WECODE Tech Fellow: Merit selected by **Harvard** to attend the largest global women in tech conference.
Stanford WIDS APAC Ambassador: High Impact Ambassador for the Stanford Global WIDS Conference.
AWS AI & ML Scholar: Awarded the AWS AI & ML Scholarship worth **\$4000**, in collaboration with Intel and Udacity.
OxML Scholar: Merit accepted to the global **Machine Learning Summer School** held at **Oxford University**.
IBM Quantum Scholar: Accepted to Qubit by Qubit's course sponsored by IBM Quantum with a **full scholarship**.
O'Reilly Scholar: Awarded the O'Reilly DEIJ Scholarship which provides unlimited access to O'Reilly learning platform.
Head of Hackerspace: Led the **largest FOSS community** on campus which focuses on technical-powered development.
Core Member at Google Developers Student Club : Organized hackathons and seminars, & advocated for technology.
Cisco Campus Ambassador : Certified for the **Cisco DevNet Associate Exam** and selected as a campus ambassador.

Publications

Empirical Study of Financial BERT Models for Sentiment Analysis and Cryptocurrency Price Correlation:
Accepted at the 9th International Conference for Convergence in Technology (I2CT) [Paper Link](#)
Exploring Influence of Music on Psychological Stress using Machine Learning and XAI Techniques [Paper Link](#)
Exploring Pivotal Dynamics of the Technological Job Market Through Data Analysis [Paper Link](#)
World of Reinforcement Learning [Paper Link](#)
Applications of Quantum Technology in Healthcare Industry [Paper Link](#)

Open Source Contributions

Kubernetes (k8s) Ecosystem Contributions

Contributor to [kubernetes-sigs/lwkd](#) (Last Week in Kubernetes Dev), authoring developer news and ecosystem updates.
Merged Pull Requests: [#802](#), [#798](#), [#795](#)